

ECO5050 AI Applications and Techniques in Economics

Lecture 3: Model Selection - Part 2

Yaein Baek

Sogang University

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Outline

- 1 Hard vs. Soft Thresholds: Shrinkage Estimator, Bagging
- 2 Properties of Model Selection Procedures
- 3 Empirical Example

Hard vs. Soft Thresholds

- By design, model selection provides hard thresholds for constructing a model in the sense that variables are either included or excluded.
- This makes sense if one is absolutely certain that the β 's are 0 or far from 0, but better forecasting models might be obtained if variables whose inclusion is uncertain are allowed to have some limited influence.
- Several approaches of soft thresholds
 - ▶ Bagging
 - ▶ Shrinkage methods
 - ▶ Combine forecasting models (forecast averaging)

Bagging

- Bagging (bootstrap aggregating) was introduced by Breiman (1996) as a method to reduce the variance of a predictor.
- The main application of Bagging in the forecasting literature has been to variable selection in prediction models; it modifies the hard threshold to soft thresholds which allows the variable selection indicators to take on a continuum of values.
- Bagging is useful when the estimator has low bias but high variance, which occurs in hard threshold model selection methods.
- The most common application of bagging is to regression trees $\Rightarrow$ Random Forest

Bagging

- The three steps for obtaining the estimator $\hat{\beta}$ from the data $\{y_{t+1}, x_t\}_{t=0}^{T-1}$
 - ① Draw resamples of the data $\{y_{t+1}, x_t\}_{t=0}^{T-1}$ to obtain a bootstrap sample $\{y_{t+1}^b, x_t^b\}_{t=0}^{T-1}$ of the same length as the original sample.
 - ② Compute an estimator of the coefficient of x_{jt} , $\hat{\beta}_j^b$, based on the b th resample of the data, $b = 1, \dots, B$.
 - ③ Compute the final estimator, $\bar{\beta}_j$, as either the mean or some other function of the resampled estimators $\{\hat{\beta}_j^b\}_{b=1}^B$.
- Steps 1 and 3 can be tailored to reflect the presumed DGP as well as the targeted properties of the final estimator.
 - ▶ If the data are dependent, such as time-series, bootstrap methods that retain the dependence are required.
 - ▶ Inoue and Kilian (2008) use a **block bootstrap** that allows for dependent data
- Step 2 chooses different models across different resamples, which translates the hard thresholds into soft thresholds.

Bagging

Block bootstrap

- Suppose we want to forecast h horizons ahead. Arrange $\{(y_{t+h}, x'_t)\}$, $t = 1, \dots, T - h$ in the form of a matrix of dimension $(T - h) \times (K + 1)$:

$$\begin{array}{cc} y_{1+h} & x'_1 \\ \vdots & \vdots \\ y_T & x'_{T-h} \end{array}$$

- Construct bootstrap samples $(y_{1+h}^b, x_{1+h}^{b'})$, $\dots$, $(y_T^b, x_T^{b'})$ by drawing with replacement blocks of m rows of this matrix, where the block size m is chosen to capture the dependence in the error term (e.g. Hall and Horowitz 1996, Gonçalves and White 2004)
 - ▶ If the forecasting model at horizon h is specified correctly in that $E(\varepsilon_{t+h}|\Omega_t) = 0$, where Ω_t denotes the date t information set, then $m = h$ (e.g. Gonçalves and Kilian 2004)
 - ▶ Otherwise, $m > h$ and data-dependent rules may be used to determine m .

Bagging

Example: Bagging used in model selection

- Suppose we use Bagging for the forward stepwise approach to obtain a forecast from a linear forecasting model with K potential predictor variables

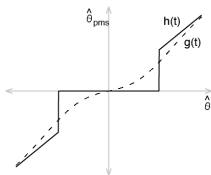
$$f_{t+1|t} = B^{-1} \sum_{b=1}^B \left(\hat{\beta}_0^b + \sum_{k=1}^K \hat{\beta}_k^b \mathbf{1}(|t_{\hat{\beta}_k}^b| > \bar{t}) x_{kt} \right)$$

- The indicator function from the b th bootstrap, $\mathbf{1}(|t_{\hat{\beta}_k}^b| > \bar{t})$, equals 1 if the absolute value of the k th t -statistic in the regression model, $t_{\hat{\beta}_k}^b$, exceeds the threshold $\bar{t}$.

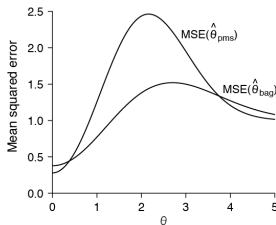
Bagging

Example: Bagging used in model selection

- $\hat{\theta}$ is the estimator obtained from the regression including all potential regressors
- Post model selection estimator: $\hat{\theta}_{pms} = h(\hat{\theta})$, where $h(t) = t\mathbf{1}(t^2 > c)$
- Bagging estimator: $\hat{\theta}_{bag} = g(\hat{\theta})$, where $g(t)$ is the smoothed transformation



(a) Selection and bagging transformations



(b) MSE of selection and bagging estimators

- Bühlmann and Yu (2002) argue that smooth transformations generally have lower variances than hard threshold; the reduction in MSE is due to a reduction in the variance of $\hat{\theta}_{bag}$

Shrinkage methods

- Shrinkage methods are a broad class of methods that reduce variance by moving an estimator $\hat{\theta}$ towards a preselected point, such as zero.
- Example: $\tilde{\theta} = (1 - w)\hat{\theta}$ for some shrinkage weight $w \in [0, 1]$. Assume $\hat{\theta} \sim (\theta, V)$

$$\text{Bias}(\tilde{\theta}) = E(\tilde{\theta}) - \theta = -w\theta$$

$$\text{Var}(\tilde{\theta}) = (1 - w)^2 V$$

$$\text{MSE}[\tilde{\theta}] = w^2 \theta' \theta + (1 - w)^2 \text{tr}(V)$$

- For the k th element, $\text{MSE}[\tilde{\theta}_k] = w^2 \theta_k^2 + (1 - w)^2 v_{kk}$
- $\text{MSE}[\tilde{\theta}_k] < \text{MSE}[\hat{\theta}_k]$ if $0 < w < 2v_{kk}/(\theta_k^2 + v_{kk})$
- $\text{MSE}[\tilde{\theta}_k]$ is minimized by the shrinkage weight $w_0 = v_{kk}/(\theta_k^2 + v_{kk})$

Shrinkage methods

- Stock and Watson (2012) show that forecasts based on some model selection methods can be viewed as generalized shrinkage methods in the context of a linear regression models with n_k predictors

$$y_{t+1} = \beta' x_t + \varepsilon_{t+1}, \quad t = 0, \dots, T-1$$

where the regressors x_t have been orthonormalized so that $X'X/T = I_{n_k}$.

- Asymptotically and under assumptions that do not require predictors to be strictly exogenous, forecasts based on pre-tests, Bagging, normal Bayes methods can be written as

$$f_{T+1|T} = \sum_{i=1}^{n_k} \psi(\kappa t_i) \hat{\beta}_i x_{iT} + o_p(1)$$

where $\hat{\beta}_i = T^{-1} \sum_{t=0}^{T-1} x_{it} y_{t+1}$ is the least square estimator of β_i , $t_i = \sqrt{T} \hat{\beta}_i / \hat{\sigma}_\varepsilon$ and $\hat{\sigma}_\varepsilon^2 = \sum_{t=0}^{T-1} (y_{t+1} - \hat{\beta}' x_t)^2 / (T - n_k)$

- e.g. For pre-testing methods $\psi(t_i) = \mathbf{1}(|\kappa| > c)$. The scalar κ depends on the forecasting method.

Consistent Selection

- A consistent model selection procedure selects the true model in large samples.
- Suppose the true model $M^* = f^*(z_t, \beta)$ is included in the set of forecasting models $\mathcal{M} = \{M_k = f_k(z_t, \beta_k), k = 1, \dots, K\}$.
- Assuming a discrete set of models, consistent model selection means that

$$P(\hat{M} = M^*) \rightarrow 1 \text{ as } T \rightarrow \infty$$

Consistent Selection

- When $\mathcal{M}_K = \{M^*, \mathcal{M}_K^u, \mathcal{M}_K^o\}$, the proof for proving consistency of model selection shows that

$$P\left(\hat{M} \in \mathcal{M}_K^u\right) \rightarrow 0 \text{ and } P\left(\hat{M} \in \mathcal{M}_K^o\right) \rightarrow 0$$

- ▶ $\mathcal{M}_K^u$: set of models that underfit (exclude variables present in the DGP)
- ▶ $\mathcal{M}_K^o$: set of models that overfit (include extraneous variables in addition to those in M^*)
- Sequential hypothesis testing is typically not a consistent model selection method and tends to overstate the size of the model.
- BIC and HQIC are consistent model selection methods; AIC and Mallows C_p are not consistent.
- v -block cross-validation is not a consistent model selection procedure when v is fixed.

Asymptotically Efficient Selection

- For linear regression models with a large or infinite number of nonzero coefficients, none of the models are correctly specified.
- The “best” model M_k^* is the model that provides a lower bound for estimation-related risk R^*
- Shibata (1980, 1981): a model selection criterion that is asymptotically efficient selects the model $\hat{M}_k^*$ that satisfies

$$\hat{R}(\hat{M}_k^*)/R^* \rightarrow 1 \text{ as } T \rightarrow \infty$$

- AIC and Mallows C_p are asymptotically efficient; BIC is not asymptotically efficient.
- IC that has deterministic penalties cannot satisfy both consistency and asymptotic efficiency (Yang, 2005; Shao 1997)

Risk for Model Selection Estimators

Monte Carlo Simulations

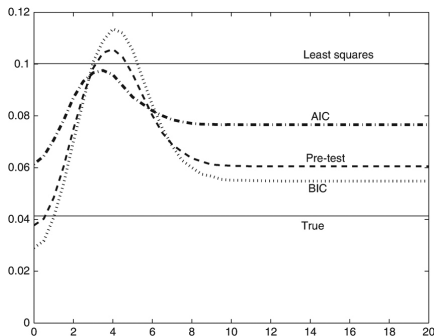
- Consistency and asymptotically efficient model selection properties are asymptotic, which means that in small sample evaluations of risk show that
 - ▶ consistent model selection methods will not generally choose the correct model with probability 1.
 - ▶ asymptotically efficient methods will not always be as good as the best model.
- Examine the risk functions of the model selection methods in Monte Carlo simulations.
- Consider forecasting y_{t+1} with $n_K = 8$ possible regressors in a linear model and assume the regressors are independent of each other.
 - ▶ Case 1: The true model has $n_{k_0} = 3$ nonzero coefficients
 - ▶ Case 2: The true model has $n_{k_0} = 7$ nonzero coefficients
- Plot the estimation-related risk (under MSE loss) of sequential hypothesis tests, BIC, and AIC.

Risk for Model Selection Estimators

Monte Carlo Simulations

- Case 1: The true model has $n_{k_0} = 3$ nonzero coefficients

$$y_{t+1} = \sum_{i=1}^{n_{k_0}} \frac{\mu}{\sqrt{n_{k_0} T}} x_{it} + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \stackrel{i.i.d.}{\sim} N(0, 1)$$

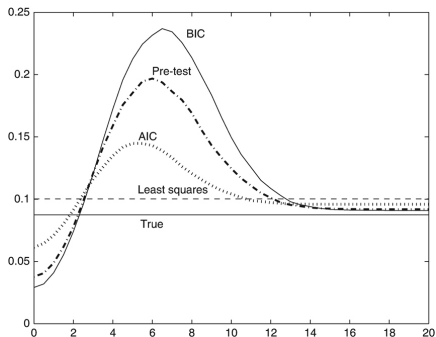


Risk for Model Selection Estimators

Monte Carlo Simulations

- Case 2: The true model has $n_{k_0} = 7$ nonzero coefficients

$$y_{t+1} = \sum_{i=1}^{n_{k_0}} \frac{\mu}{\sqrt{n_{k_0} T}} x_{it} + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \stackrel{i.i.d.}{\sim} N(0, 1)$$



Empirical Example

Forecasting stock returns (Welch and Goyal, 2008)

- We use a linear forecasting model $y_{t+1} = x_t' \beta + \varepsilon_{t+1}$ to forecast stock market returns.
- y_{t+1} = quarterly returns on a value-weighted portfolio of US stocks measured in excess of a three-month T-bill rate
- Total 11 potential predictor variables x_t
 - ▶ log dividend-price ratio (dp), log dividend yield (dy), log earnings-price ratio (ep), book-to-market ratio (bm), net equity issues (ntis), three-month T-bill rate (tbl), long term returns (ltr), term spread (tms), default yield (dfy), inflation (infl), investment-capital ratio (ik)
- Data sample: 1947Q1-2013Q4

Empirical Example

Forecasting stock returns

- We generate pseudo out-of-sample forecasts using a recursively expanding estimation window
 - ▶ Allows us to simulate the forecasting models' "real-time" performance, we can see how much the selected models change over time and across different economic environments
 - ▶ Forecast for 1970Q1 is generated from a model using data from 1947Q1-1969Q4; the forecast for 1970Q2 is generated from a model reestimated from data 1947Q1-1970Q1 and so on.



Source: www.uber.com/en-KR/blog/omphalos/

Empirical Example

Forecasting stock returns

- Consider linear forecasting models and use the following model selection methods:
 - ▶ Forward stepwise approach: maximum p -value of 0.05 for a predictor to be included
 - ▶ Backward stepwise approach: minimum p -value of 0.10 for a predictor to be removed
 - ▶ Lasso: Penalty $\lambda = 2c\sigma\sqrt{T}\Phi^{-1}(1 - \alpha/2n_k)$, where c is $c_1 = 0.002$ or $c_2 = 0.006$
 - ▶ Bagging
 - ▶ Cross-validation: evaluation sample of $T_v = 0.5(0.5T - \sqrt{T} - 1)$ observations for a sample of size T
- Benchmark model: the prevailing mean, forecast for y_{t+1} is $t^{-1} \sum_{s=1}^t y_s$

Empirical Example

Forecasting stock returns

TABLE 6.1:

Correlations among forecasts (shown below the diagonal) and forecast errors (shown above the diagonal) for different approaches to forecasting quarterly US stock market returns.

	<i>AIC</i>	<i>BIC</i>	<i>Forward step</i>	<i>Backward step</i>	<i>Bagging $\alpha = 1\%$</i>	<i>Bagging $\alpha = 5\%$</i>	<i>Lasso $c = 2$</i>	<i>Lasso $c = 6$</i>	<i>Cross valid</i>	<i>Prevailing mean</i>
AIC	1.0000	0.9689	0.9629	0.9611	0.9594	0.9746	0.9600	0.9505	0.9362	0.9313
BIC	0.7529	1.0000	0.9656	0.9592	0.9608	0.9679	0.9681	0.9640	0.9389	0.9382
Forward step	0.6859	0.6899	1.0000	0.9737	0.9664	0.9711	0.9738	0.9701	0.9468	0.9469
Backward step	0.6870	0.6557	0.7624	1.0000	0.9578	0.9638	0.9834	0.9546	0.9434	0.9383
Bagging $\alpha = 1\%$	0.7135	0.6583	0.6240	0.5739	1.0000	0.9924	0.9583	0.9778	0.9754	0.9897
Bagging $\alpha = 5\%$	0.7915	0.6884	0.6723	0.6349	0.9758	1.0000	0.9662	0.9675	0.9636	0.9662
Lasso $c = 2$	0.6834	0.7343	0.7708	0.8619	0.6281	0.6781	1.0000	0.9615	0.9387	0.9359
Lasso $c = 6$	0.5146	0.6441	0.6481	0.5039	0.3659	0.4254	0.6197	1.0000	0.9716	0.9768
Cross valid	0.3261	0.3038	0.2992	0.3451	0.2581	0.3409	0.3103	0.3772	1.0000	0.9778
Prevailing mean	-0.1019	0.0821	-0.0974	-0.1150	-0.0150	-0.0764	-0.0771	0.1346	0.1167	1.0000

Empirical Example

Forecasting stock returns

TABLE 6.2:

Out-of-sample forecasting performance for different forecasting methods applied to quarterly US stock market returns. Std and Bias are the standard deviation and bias of the recursive forecasts, while RMSE and R^2 are the root mean squared forecast error and the R^2 , the latter measured relative to the prevailing mean model. DM lists the Diebold–Mariano test for equal predictive accuracy. Sign and PT-test give the percentage of correctly predicted signs of quarterly excess returns and the associated value of the Pesaran–Timmermann (1992) test statistic, along with its p -value.

<i>Method</i>	<i>Std(%)</i>	<i>Bias(%)</i>	<i>RMSE(%)</i>	<i>R²(%)</i>	<i>DM stats</i>	<i>Sign(%)</i>	<i>PT-test</i>	<i>p-val</i>
AIC	3.3089	−0.0324	9.1511	−14.4390	−1.9679	55.1136	−0.0953	0.4620
BIC	3.1860	0.2059	9.1796	−15.1516	−2.4083	54.5455	0.0210	0.4916
Forward step	2.8103	0.0343	8.7978	−5.7728	−1.1157	53.4091	−0.9605	0.1684
Backward step	3.0699	0.2602	8.9566	−9.6253	−1.4700	51.7045	−0.2834	0.3884
Bagging $\alpha = 1\%$	1.2097	−0.1898	8.5683	−0.3265	−0.1366	59.6591	−0.3509	0.3628
Bagging $\alpha = 5\%$	2.2270	0.1921	8.7244	−4.0162	−0.8453	52.2727	−0.6508	0.2576
Lasso c_1	3.1257	0.0511	8.9145	−8.5968	−1.4748	54.5455	0.0210	0.4916
Lasso c_2	1.8825	−0.1747	8.7124	−3.7303	−0.9242	60.2273	0.0304	0.4879
Cross valid	1.8569	0.3547	8.7983	−5.7842	−1.2595	57.3864	0.4494	0.3266
Prevailing mean	0.1944	−0.4041	8.5543	0.0000	0.0000	61.9318	NaN	NaN

Empirical Example

Forecasting stock returns

- The root mean squared error (RMSE) and R^2 measures the model selection method's predictive accuracy.
- The RMSE suggests that the model selection methods associated with the least volatile forecasts perform best, suggesting that predictors are not very powerful and parameter estimation error is important in this application.
- The out-of-sample R^2 measures relative to the prevailing mean model. Negative values show underperformance relative to the prevailing mean forecast.
- Forecast evaluation methods will be discussed in detail later on.

Empirical Example

Forecasting stock returns

